

ZIKUN LI

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EDUCATION

School of Computer Science, Carnegie Mellon University (CMU)

Doctoral student in Computer Science

Master of Science in Computer Science

Pittsburgh, PA, US

Aug 2022 - Expected 2027

Aug 2026

- Advised by Prof. Zhihao Jia
- Research interests: Efficient LLM Architectures, LLM Training, LLM Inference, Reinforcement Learning

School of Electronic Engineering and Computer Science, Peking University (PKU)

Bachelor of Science in Computer Science

Beijing, China

Sep 2017 - Jul 2021

PUBLICATIONS

1. **Zikun Li***, Yixuan Mei*, Shiqi Pan, Zixuan Chen, Xiaowen Zhang, Mengdi Wu, Shuhuai Lin, Yutong Yang, Zhihao Zhang, Xupeng Miao, Rashmi Vinayak, and Zhihao Jia, “*OpWeave: Flexible Operator Disaggregation for Heterogeneous LLM Serving*”, arXiv preprint arXiv:2609.14237, 2026 (* indicates equal contribution). **[LLM Inference]**
2. **Zikun Li***, Zhuofu Chen*, Remi Delacourt, Gabriele Oliaro, Zeyu Wang, Qinghan Chen, Shuhuai Lin, April Yang, Zhihao Zhang, Zhuoming Chen, Sean Lai, Xinhao Cheng, Xupeng Miao, and Zhihao Jia, “*AdaServe: Accelerating Multi-SLO LLM Serving with SLO-Customized Speculative Decoding*”, ACM European Conference on Computer Systems (EuroSys), 2026 (* indicates equal contribution). **[Speculative Decoding]**
3. Yixuan Mei, **Zikun Li**, Zixuan Chen, Shiqi Pan, Mengdi Wu, Xupeng Miao, Zhihao Jia, and K. V. Rashmi, “*Coral: Cost-Efficient Multi-LLM Serving over Heterogeneous Cloud GPUs*”, arXiv preprint arXiv:2605.04357, 2026. **[LLM Inference]**
4. Lijie Yang, Zhihao Zhang, Zhuofu Chen, **Zikun Li**, and Zhihao Jia, “*TidalDecode: Fast and Accurate LLM Decoding with Position Persistent Sparse Attention*”, International Conference on Learning Representations (ICLR), 2025. **[Sparse Attention]**
5. **Zikun Li**, Jinjun Peng, Yixuan Mei, Sina Lin, Yi Wu, Oded Padon, and Zhihao Jia, “*Quarl: A Learning-Based Quantum Circuit Optimizer*”, ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA), 2024. **[Reinforcement Learning]**
6. Ruijie Miao, Zheng Zhong, Jiarui Guo, **Zikun Li**, Tong Yang, and Bin Cui, “*BurstSketch: Finding Bursts in Data Streams*”, IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.
7. Mingkuan Xu, **Zikun Li**, Oded Padon, Sina Lin, Jessica Pointing, Auguste Hirth, Henry Ma, Jens Palsberg, Alex Aiken, Umut A. Acar, and Zhihao Jia, “*Quartz: Superoptimization of Quantum Circuits*”, ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI), 2022.
8. Zheng Zhong*, Shen Yan*, **Zikun Li***, Decheng Tan, Tong Yang, and Bin Cui, “*BurstSketch: Finding Bursts in Data Streams*”, ACM SIGMOD International Conference on Management of Data (SIGMOD), 2021 (* indicates equal contribution).
9. Jizhou Li*, **Zikun Li***, Yifei Xu*, Shiqi Jiang, Tong Yang, Bin Cui, Yafei Dai, and Gong Zhang, “*WavingSketch: An Unbiased and Generic Sketch for Finding Top-k Items in Data Streams*”, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2020 (* indicates equal contribution).

EXPERIENCE

Understanding Sparse Attention in LLMs through Controlled Experiments

Menlo Park, CA

Research Scientist Intern, MSL FAIR at Meta, Manager: Josh Fromm

Jun 2026 - Aug 2026

- Built oracle attention-scoring kernels and integrated sparse attention into SGLang and TorchTitan for controlled inference and training experiments.
- Compared dense and oracle-sparse attention across 1.7B–32B models, finding near-dense simple retrieval performance but persistent deficits on an aggregation task.
- Identified longer reasoning traces and more frequent truncation under sparse attention on AIME26.
- Conducted matched dense-versus-sparse continual-pretraining experiments, showing that average loss recovery can mask persistent long-context deficits.

Operator-Disaggregated LLM Serving over Heterogeneous GPUs

Pittsburgh, PA (Remote with ByteDance Seed)

Student Researcher, ByteDance Seed-Training-Infra & CMU, Mentor: Dr. Ziheng Jiang

Sep 2025 - Dec 2025

- Studied emerging **operator-disaggregated** LLM serving, quantifying its benefits on throughput and cost and identifying optimal disaggregation strategies.
- Developed a search algorithm that, given an LLM and heterogeneous GPUs, outputs the optimal end-to-end serving plan (operator partitioning, parallelization strategy, device assignment, and batch sizing).
- Built a distributed serving runtime on **vLLM** and **NCCL** to enable flexible operator-disaggregated execution across heterogeneous GPUs.
- Preliminary evaluation shows up to **30% cost reduction** under latency SLO requirements.

Collaborative CPU–Accelerator Control Flow for MoE Serving on Dataflow Hardware

Santa Clara, CA

Applied Scientist Intern, AWS Neuron Science Team, Mentor: Dr. Ziyang Xu

May 2025 - Aug 2025

- Studied **dynamic control flow challenges in MoE inference** on dataflow accelerators where expert routing makes block counts runtime-dependent.
- Designed a **CPU–accelerator co-scheduling runtime** that offloads routing decisions to CPU and launches expert kernels asynchronously.
- Proposed a **branch-delay-slot scheduling technique** to overlap gating-data transfer with lower-bound expert execution.
- Implemented **multi-block kernels** and **async launches** to restore cross-block optimizations and hide launch latency.
- Achieved up to **1.3× speedup** over static scheduling, eliminating padding-block computation and reducing launch stalls **64μs → 27μs**.

Robust Loop Scheduler in Machine Learning Compiler for Trainium

Santa Clara, CA

Applied Scientist Intern, AWS Neuron Science Team, Mentor: Dr. Ziyang Xu

May 2024 - Dec 2024

- Studied and verified the source of uncertainties in the latency of the scheduling units.
- Designed and implemented a robust loop scheduler that is resistant to uncertainties.
- Achieved up to **1.46× speedup** on attention kernels with the proposed algorithm.

TEACHING EXPERIENCE

Teaching assistant 15-442/642 Machine Learning Systems

Jan 2026 - May 2026

Teaching assistant 15-418/618 Parallel Computer Architecture and Programming

Aug 2023 - Dec 2023

SERVICE

Program Committee Member, MLSys 2026

SELECTED TECHNICAL SKILLS

Python, C++, CUDA, Triton; PyTorch, TorchTitan, vLLM, SGLang, NCCL; NVIDIA Nsight Systems/Compute